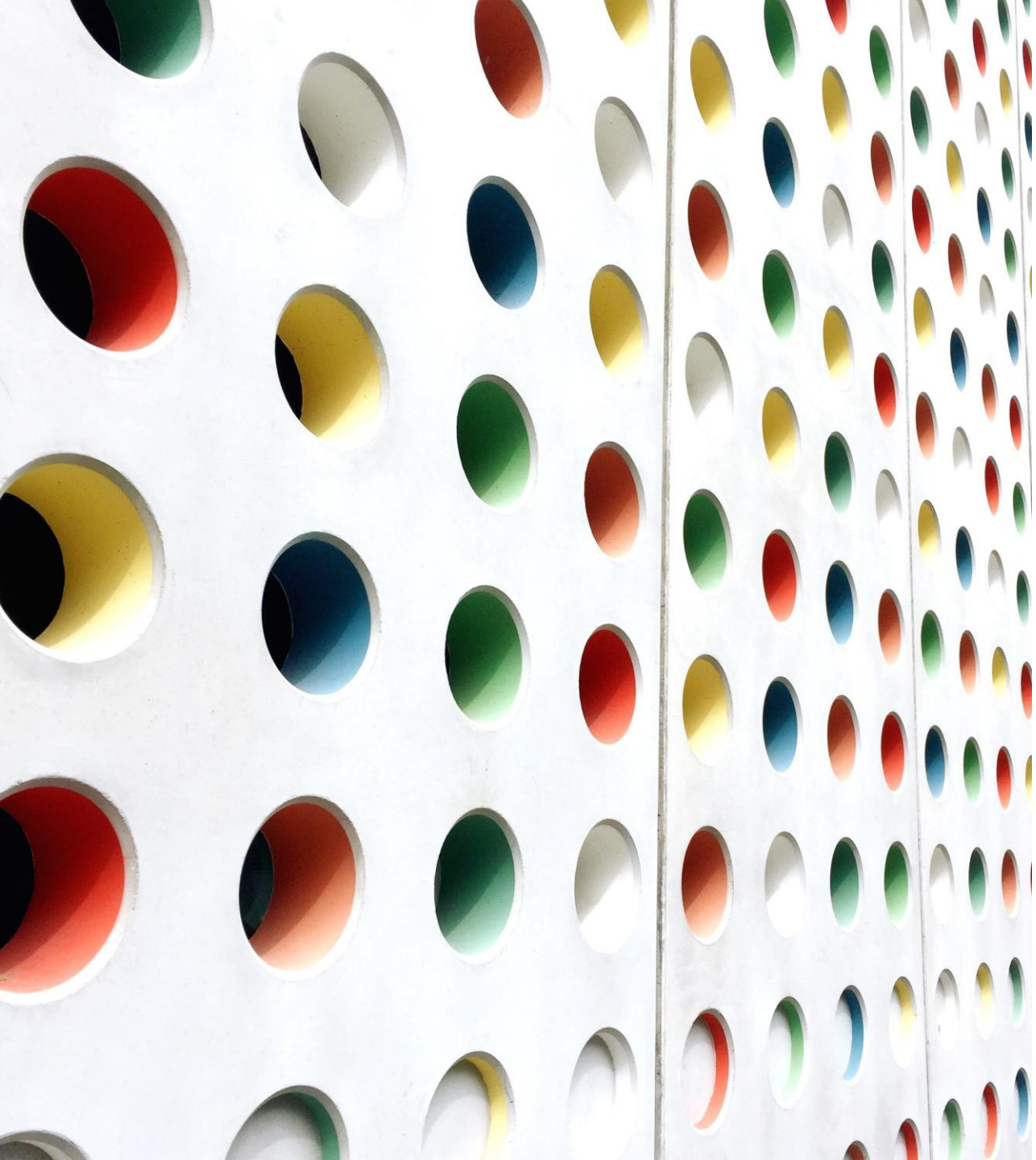


“Why Should I Trust You?” Explaining the Predictions of Any Classifier

Presented by Ritika Verma

XAI reading group: Session 2

Motivation

Modern Machine Learning(ML) models are **black boxes.**

Users must decide:

- Can I **trust this prediction**?
- Can I **trust this model**?

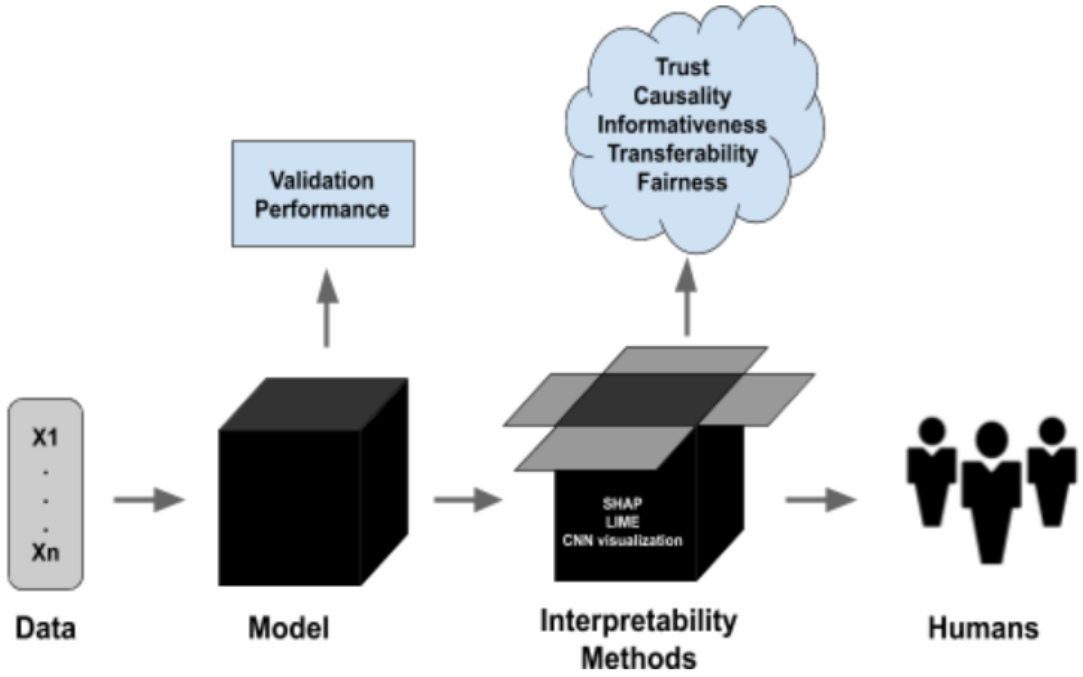
What does "trust" mean?

Trusting a prediction

- Should I act on this single output?

Trusting a model

- Should I deploy this model in the real world?



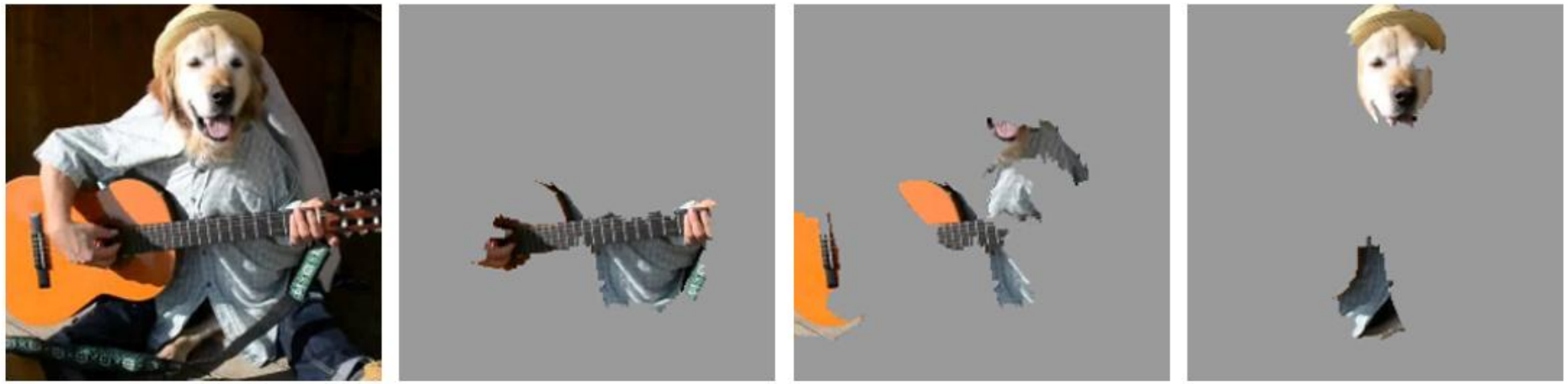
Husky or Wolf?

(a) Husky classified as wolf (b) Explanation

The model detects snow instead of Husky for the classification task. It cannot be trusted.

Explaining an Image Classification Problem

The model detects features correctly for classification. It can be trusted.



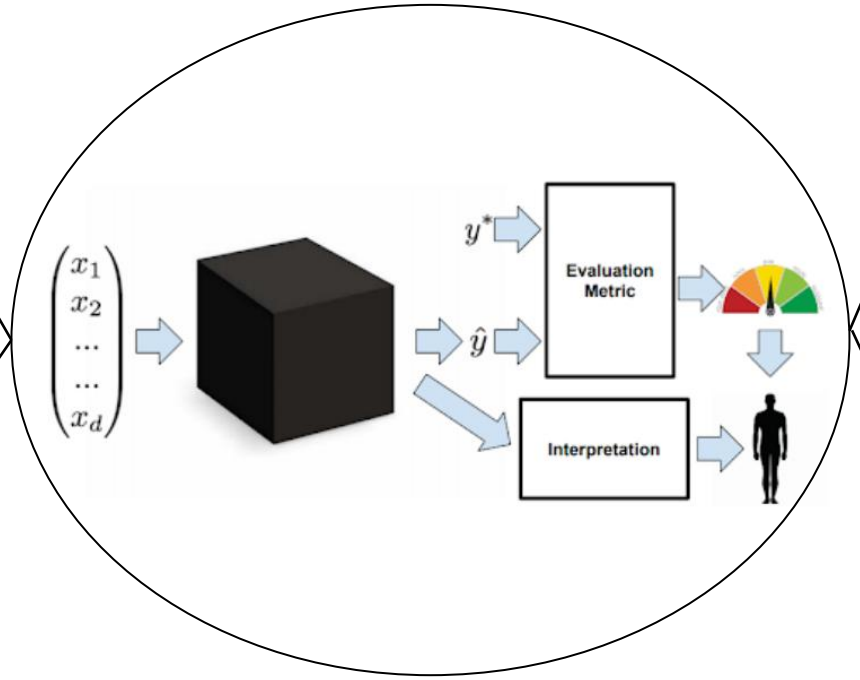
(a) Original Image (b) Explaining *Electric guitar* (c) Explaining *Acoustic guitar* (d) Explaining *Labrador*

Figure 4: Explaining an image classification prediction made by Google's Inception neural network. The top 3 classes predicted are "Electric Guitar" ($p = 0.32$), "Acoustic guitar" ($p = 0.24$) and "Labrador" ($p = 0.21$)

Core Idea of the Paper

Explain individual predictions (LIME).

Explain predictions to build trust.



Select representative explanations to understand the model globally (SP-LIME).

Must work for any classifier (model-agnostic).

Desired Properties of Explanations

1) **Interpretability**

2) **Local fidelity**

3) **Model-agnostic**

4) **Global understanding**

What Is LIME?

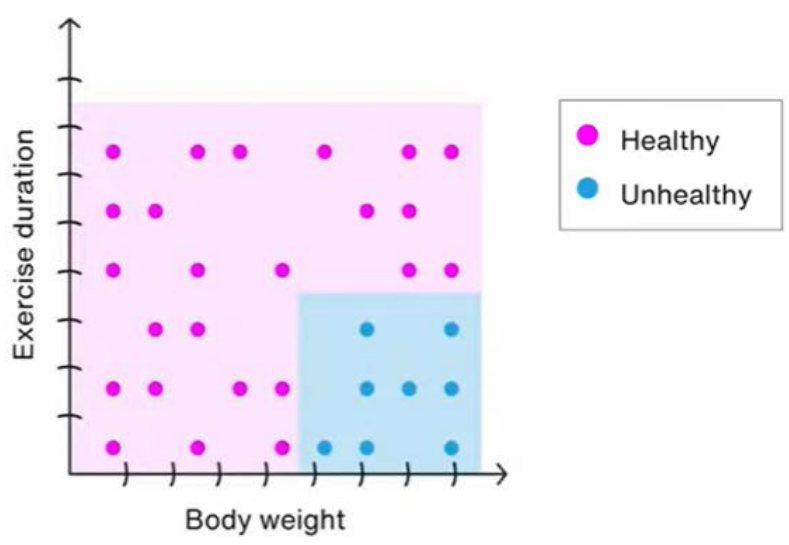
Local → Local neighbourhood of the instance

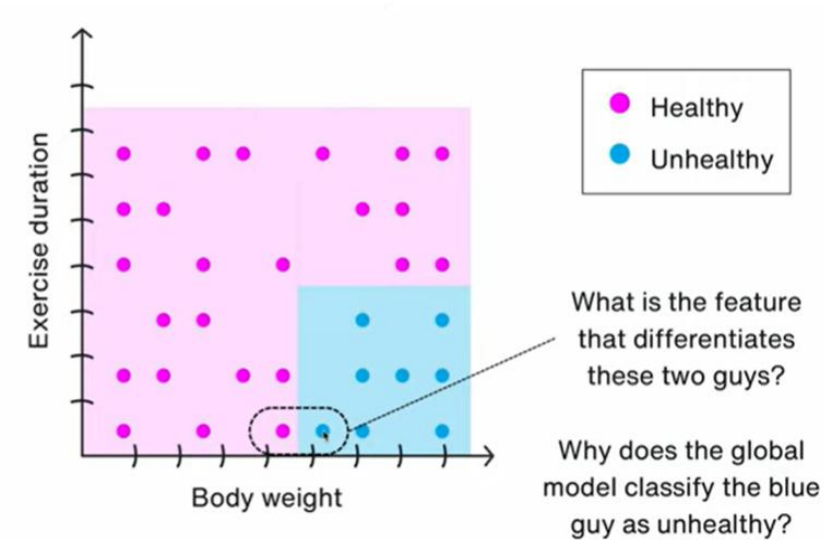
Interpretable → A human should be able to interpret

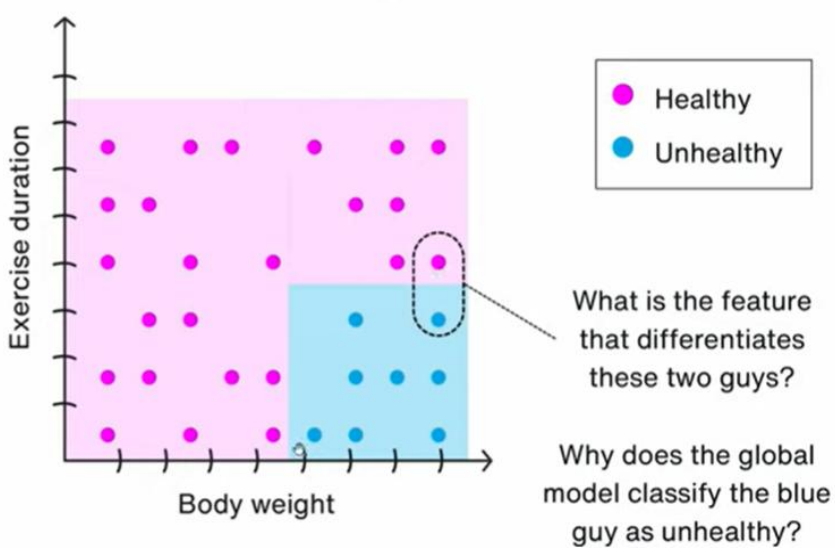
Model-agnostic → Applicable for all models

Explanations → Explanation that helps the interpretation

Global and Local Interpretability







Text Classification

I have a medical emergency. Hence won't be able to attend the meeting today.	Important
Hi may I get the information about your service?	Not important

I have a medical emergency. Hence won't be able to attend the meeting today.	Important
Hi may I get the information about your service?	Not important

Intuition Behind LIME

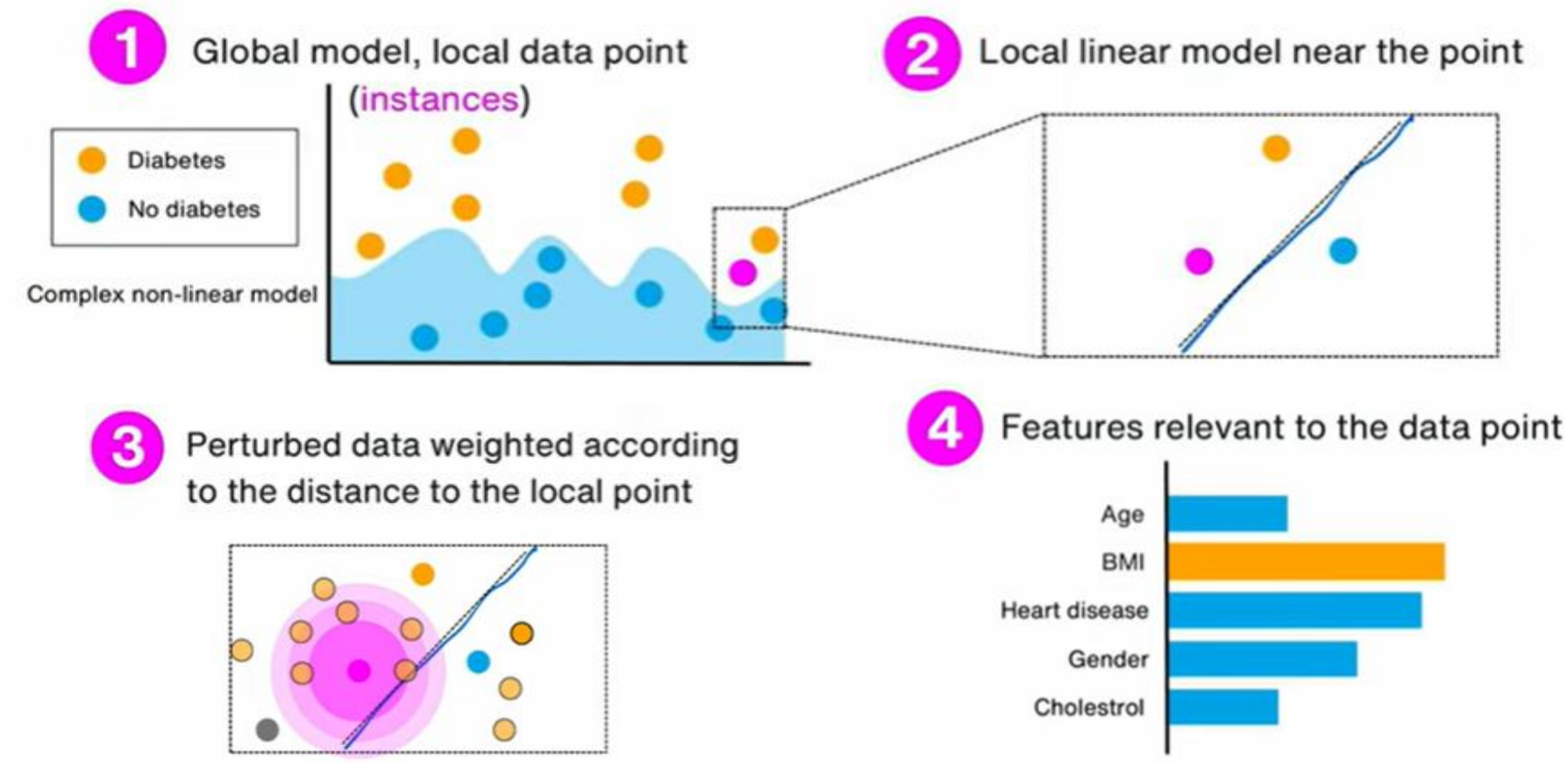
Linear models are readily interpretable.

For example, if we have:

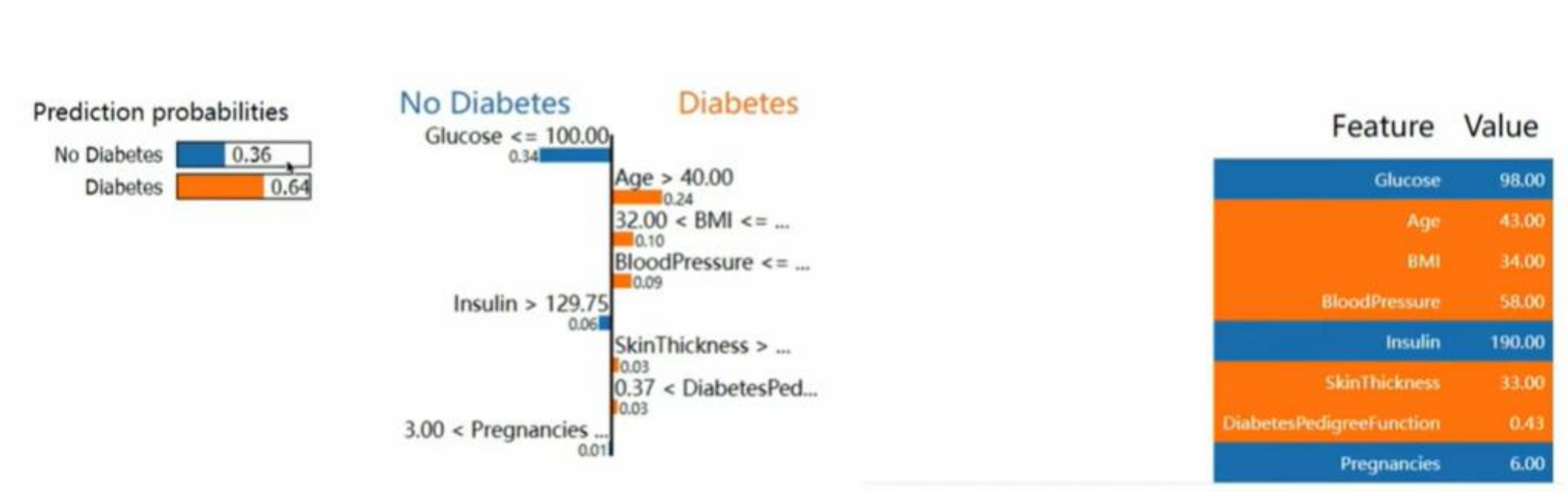
$$y = w_1 x_1 + w_2 x_2 + w_3 x_3$$

w1 is the contribution to the prediction of the feature x1 for the specific data instance and high value means a high contribution.

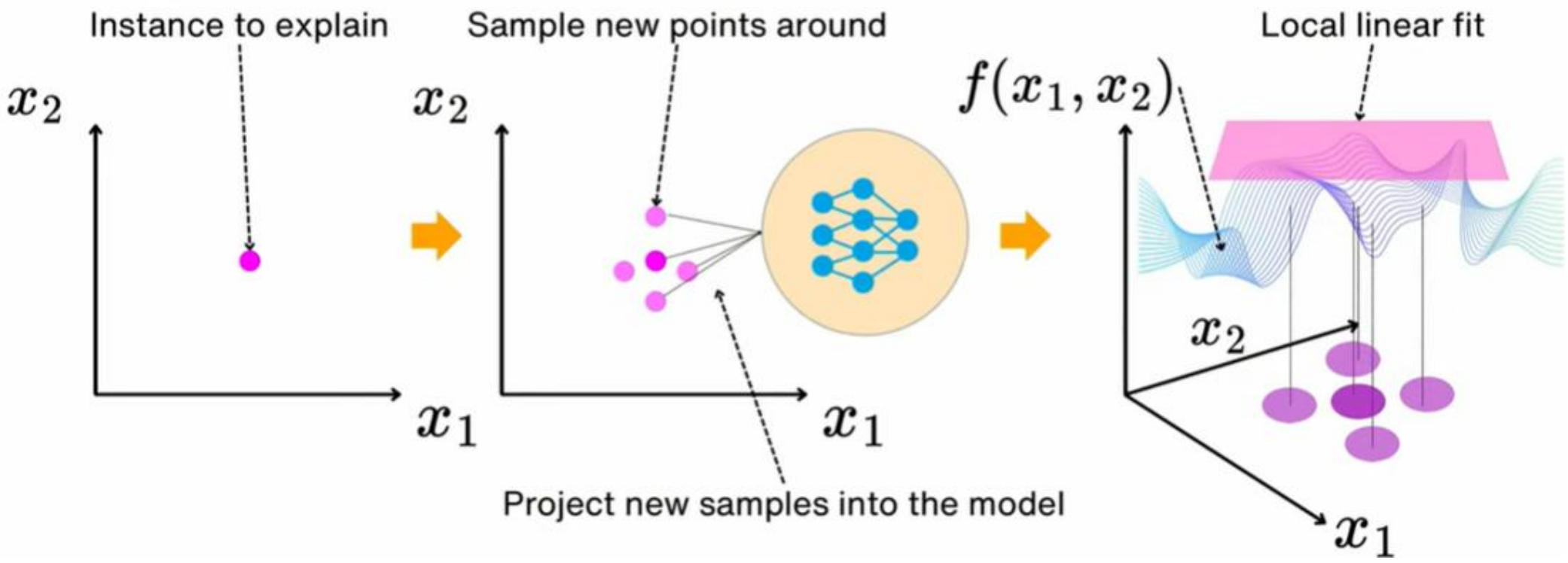
LIME Implementation in 4 Steps



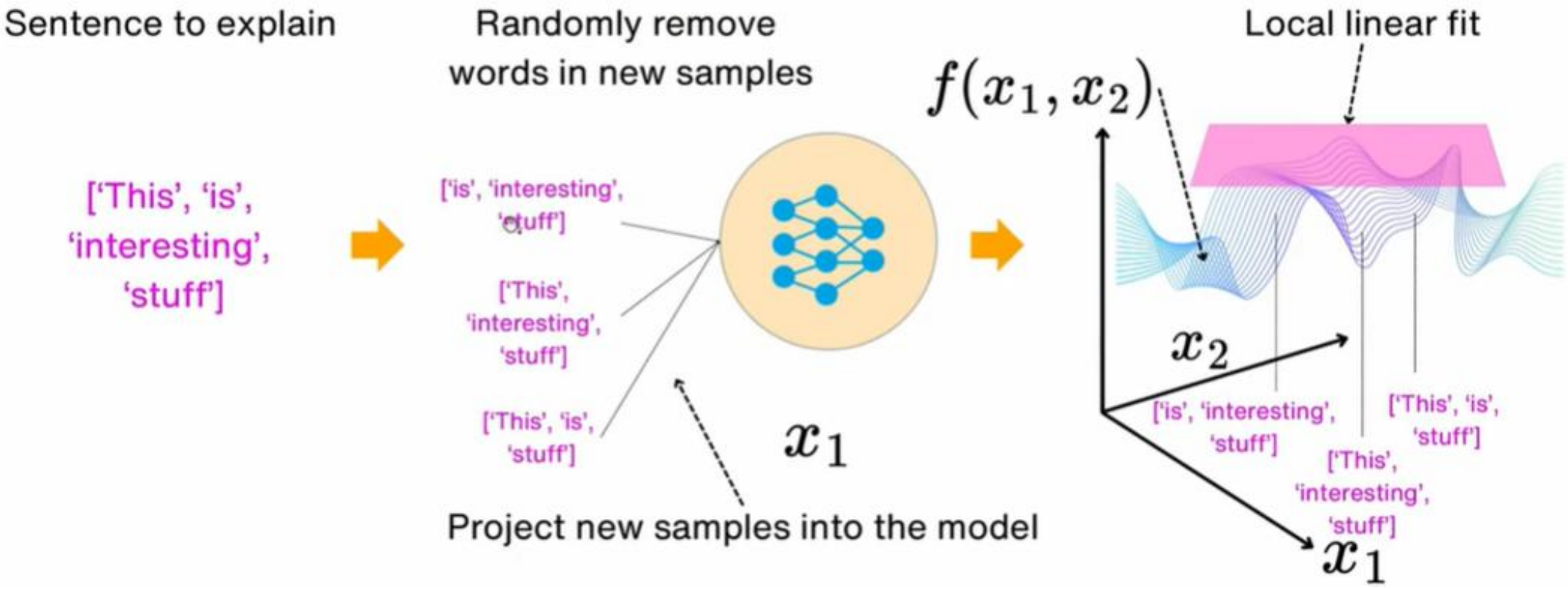
LIME Prediction



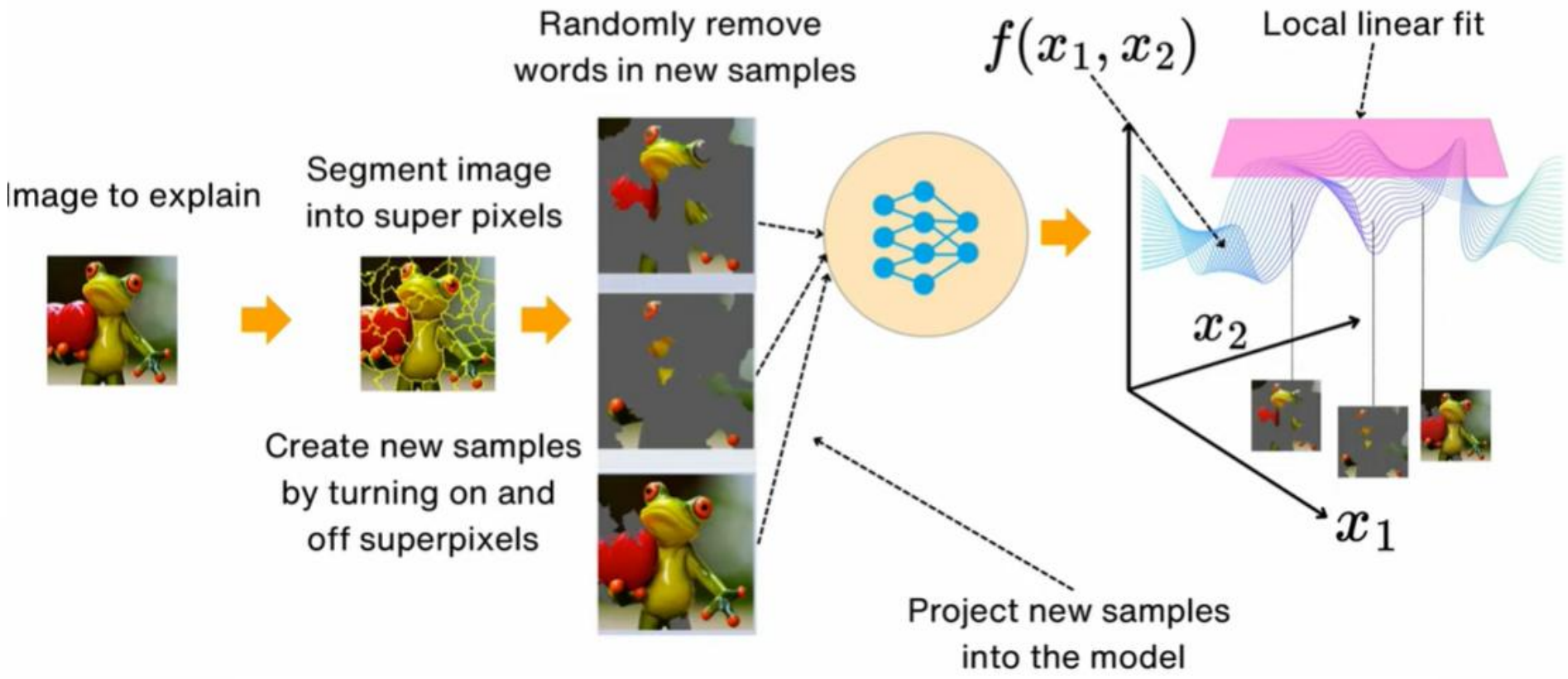
LIME with Tabular Data



LIME with Text Data



LIME with Image Data



Mathematical explanation for LIME

Notation Overview

$f: \mathbb{R}^d \rightarrow \mathbb{R}$:original black-box model

$x \in \mathbb{R}^d$:instance being explained

$x' \in \{01\}^{d'}$, interpretable representation

$g \in G$: explanation model

Interpretable Representations

Model features ≠ explanation features

Examples:

Text

- Model: embeddings
- Explanation: word presence (0/1)

Images

- Model: pixels
- Explanation: super-pixels (present/absent)

Fidelity-Interpretability Trade-off

We have already discussed that Explanations must be:

- Interpretable to humans
- Faithful to the model locally

These goals conflict for complex models.

LIME explicitly balances this trade-off, **HOW**?

Optimization Problem

$$\xi(x) = \arg\min_{g \in G} \overset{1}{\boxed{\mathcal{L}(f, g, \pi_x)}} + \overset{2}{\boxed{\Omega(g)}}$$

Predicted instance we want to explain

Family of interpretable models

Complex model being explained

Simple/surrogate model

Neighbourhood of x

Complexity measure of g

We have two loss functions marked with 1 and 2.

Optimization problem returns that **g** which will minimize both 1 and 2.

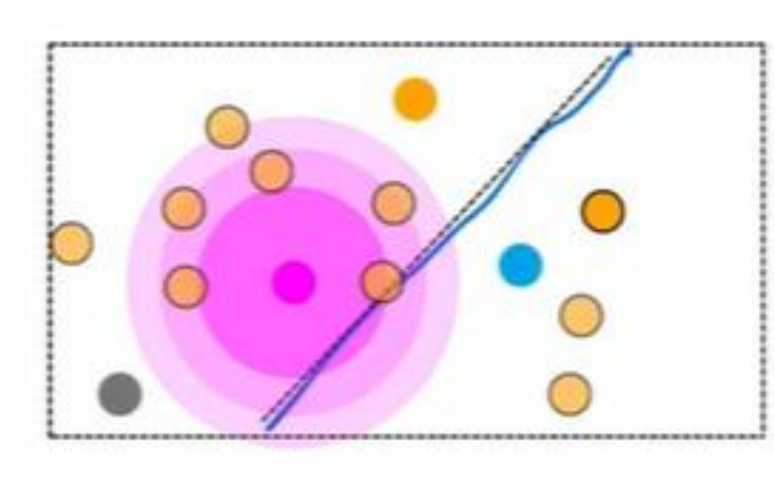
BUT WHY DO WE NEED TO MINIMIZE?

WHY Minimize?

$$\mathcal{L}(f, g, \pi_x)$$

This is Local Loss function "L".

We want to know the local behavior of L in the vicinity of x.
Thus, we generate perturbation in x.



$$\mathcal{L}(f, g, \pi_x) = \sum_{z \in \mathcal{Z}} \pi_x(z) \left(f(z) - g(z)\right)^2$$

Complex model prediction

Simple/surrogate model prediction

$$\pi_x(z) = exp(-D(x, z)^2/\sigma^2)$$

Euclidean distance

Complexity penalty

To ensure **interpretability**, explanations are constrained to use **at most** K **features**.

$$\Omega(g) = \infty \mathbb{1}[\|w_g\|_0 > K]$$

This constraint rejects explanations that involve more than K non-zero weights, reflecting human cognitive limits.

Therefore, L is minimized to make the **model faithful** and "$\Omega(g)$" needs to be minimum to make the **model interpretable.**

LIME Algorithm

Algorithm 1 Sparse Linear Explanations using LIME

Require: Classifier f, Number of samples N
Require: Instance x, and its interpretable version x'
Require: Similarity kernel π_x, Length of explanation K

```
Z ← {}
for i ∈ {1, 2, 3, ..., N} do
    z'_i ← sample_around(x')
    Z ← Z ∪ ⟨z'_i, f(z_i), π_x(z_i)⟩
end for
w ← K-Lasso(Z, K)   ▷ with z'_i as features, f(z) as target
return w
```

1. Sample N perturbed instances around interpretable input x'
2. Query black-box model f on each perturbed instance
3. Weight samples using proximity kernel π_{x}
4. Fit a sparse linear model using K-LASSO
5. K-LASSO is a 2-Step process. (Lasso method to select K feature and learning the weights via Least Squares.)
6. Return feature weights as the local explanation.

Submodular Pick-LIME (Global Understanding)

Explaining a single prediction is insufficient to assess trust in a model.

Users have limited time to inspect explanations.

Need a **small, representative set of explanations** to understand global model behavior.

Goal: select explanations that are informative and non-redundant.

Note: Given a set of instances **X** and a user budget **B,** i.e. a subset V of B. Selected explanations should collectively represent the model's behavior. This selection step is referred to as the **pick step.**

Explanation Matrix and Feature Importance

Construct explanation matrix $W \in \mathbb{R}^{n \times d'}$

- Rows: instances
- Columns: interpretable features
- W_{ij} :**importance of feature j in explanation of instance i.**

Define global feature importance:

$$I_j = \sqrt{\sum_{i=1}^{n} W_{ij}}$$

Features explaining many instances receive higher importance

Coverage Function

1) Coverage measures how well a set of explanations represents important features

$$c(V, W, I) = \sum_{j=1}^{d'} \mathbf{1}\left[\exists i \in V : W_{ij} > 0\right] \cdot I_j$$

2) Each feature contributes **once**, regardless of repetition

3) Encourages diversity and avoids redundant explanations

Submodular Pick Optimization

Pick problem:

$$Pick(\mathcal{W}, I) = \underset{V, |V| \leq B}{\operatorname{argmax}}\, c(V, \mathcal{W}, I)$$

1) Objective is NP-hard but **submodular**

2) Sub modularity implies diminishing returns

3) Greedy selection yields a $1 - 1/e$ approximation guarantee.

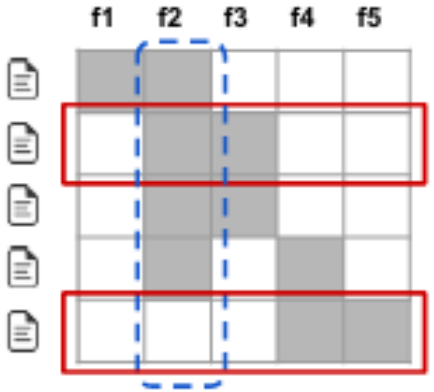


Figure 5: Toy example $\mathcal{W}$. Rows represent instances (documents) and columns represent features (words). Feature f2 (dotted blue) has the highest importance. Rows 2 and 5 (in red) would be selected by the pick procedure, covering all but feature f1.

Algorithm 2: Submodular Pick (SP-LIME)

Algorithm 2 Submodular pick (SP) algorithm

Require: Instances X, Budget B

for all $x_i \in X$ **do**

  $\mathcal{W}_i \leftarrow$ **explain**(x_i, x_i') ▷ Using Algorithm 1

end for

for $j \in \{1 \dots d'\}$ **do**

  $I_j \leftarrow \sqrt{\sum_{i=1}^{n} |\mathcal{W}_{ij}|}$ ▷ Compute feature importances

end for

$V \leftarrow \{\}$

while $|V| < B$ **do** ▷ Greedy optimization of Eq (4)

  $V \leftarrow V \cup \text{argmax}_i \, c(V \cup \{i\}, \mathcal{W}, I)$

end while

return V

1. Generate LIME explanations for all instances.
2. Compute explanation matrix W and feature importance I
3. Iteratively add the instance with maximum marginal coverage gain.
4. Output V non-redundant explanations for global model inspection.

References

1) https://ishwaryasriraman.medium.com/from-opaque-to-transparent-understanding-machine-learning-models-with-lime-3f2a2d147642

2) https://towardsdatascience.com/interpreting-image-classification-model-with-lime-1e7064a2f2e5/

3) https://www.youtube.com/watch?v=d6j6bofhj2M